

Day 03: Measures of Central Tendency

Understanding the Center of Data

100 Days of Data Science

Why Central Tendency is Important

In Data Science, we often want a single value that represents the whole dataset. Measures of central tendency help summarize data in a simple and meaningful way.

They answer the question: **Where does most of the data lie?**

Main Measures of Central Tendency

The three main measures are:

- Mean
- Median
- Mode

Each measure has different use cases depending on data distribution.

Mean (Arithmetic Mean)

The mean is the average of all values in a dataset.

Formula

$$\text{Mean} = \frac{\sum x_i}{n}$$

Intuition

Mean represents the value each observation would have if the total were equally distributed.

Example

Data: 60, 70, 80, 90, 100

$$\text{Mean} = \frac{60 + 70 + 80 + 90 + 100}{5} = 80$$

Advantages

- Easy to calculate
- Uses all data points

Limitations

- Highly affected by outliers

Median

The median is the middle value when data is arranged in order.

Intuition

Median splits the dataset into two equal halves.

Example

Data: 20, 22, 25, 30, 500

Median = 25

Advantages

- Not affected by extreme values
- Best for skewed data

Mode

Mode is the value that occurs most frequently.

Example

Data: 7, 8, 9, 9, 9, 10

Mode = 9

Use Cases

- Most common category
- Popular product size or choice

Comparison of Mean, Median and Mode

Feature	Mean	Median	Mode
Affected by Outliers	Yes	No	No
Best for Skewed Data	No	Yes	Sometimes
Used for Categories	No	No	Yes

Which One Should You Use?

- Use **Mean** for symmetric data without outliers
- Use **Median** when outliers exist
- Use **Mode** for categorical data

Role in Data Science

- Mean used as baseline in ML models
- Median used in income and housing data
- Mode used in classification problems

Final Takeaway

There is no single best measure of center. The correct choice depends on the nature of data. Understanding this helps make better data-driven decisions.